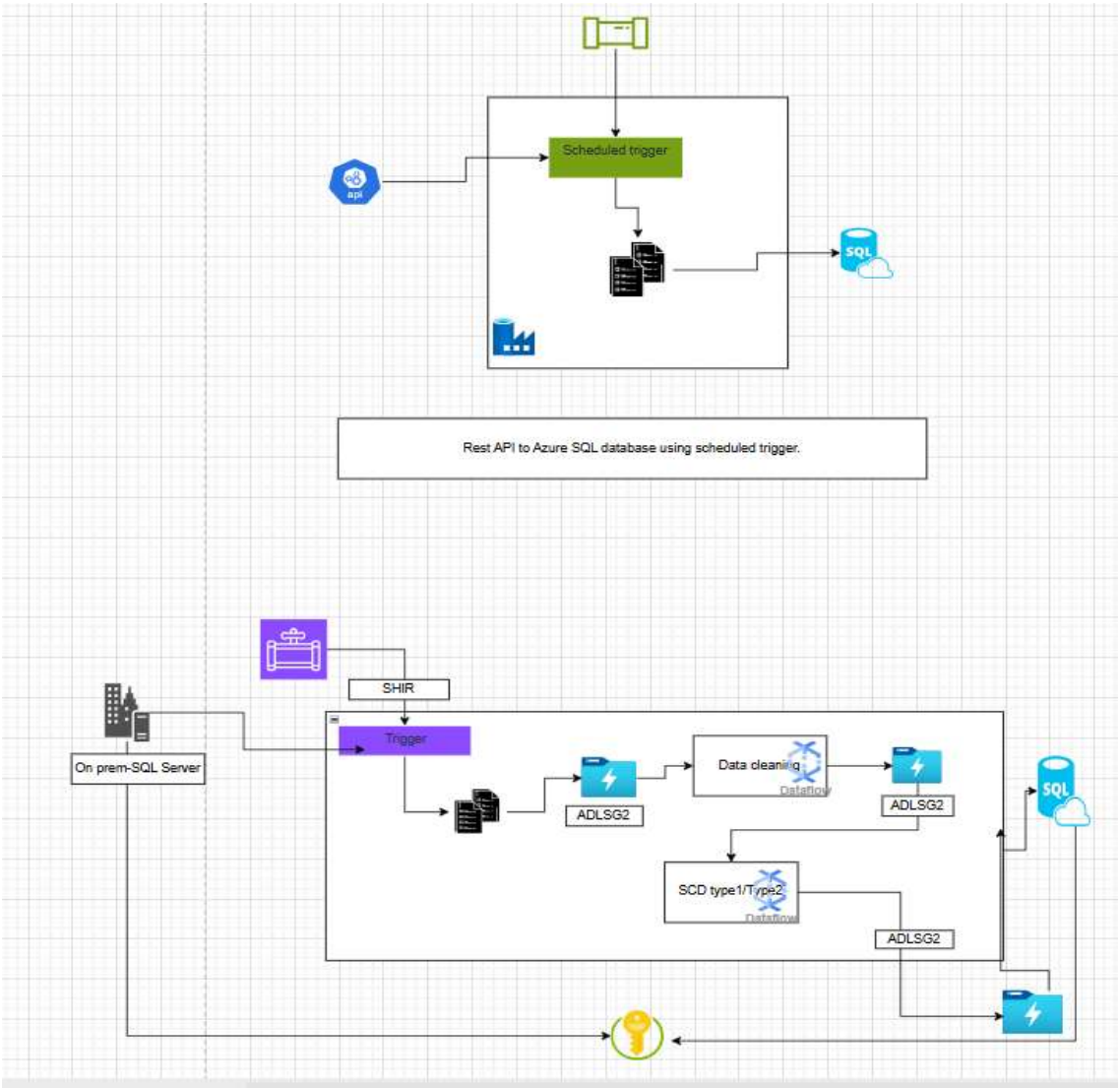
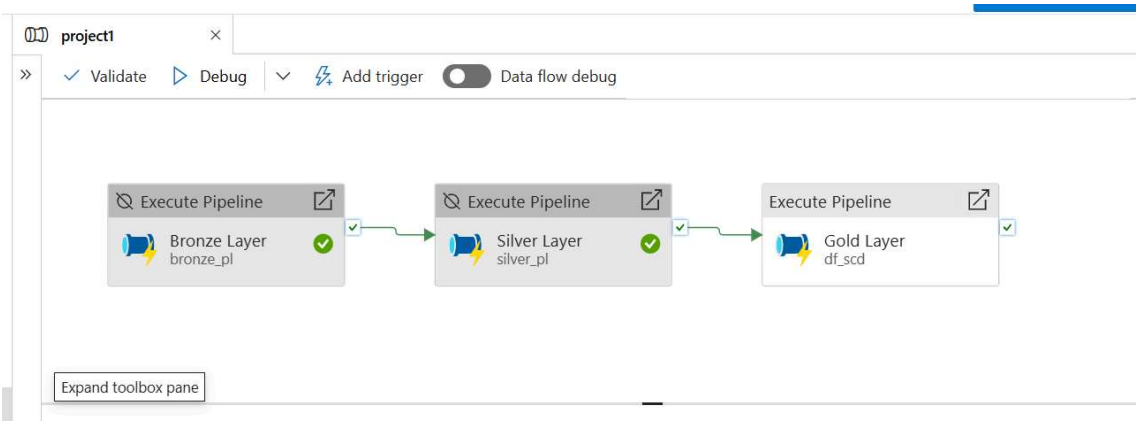
Project-1

Architecture diagram:



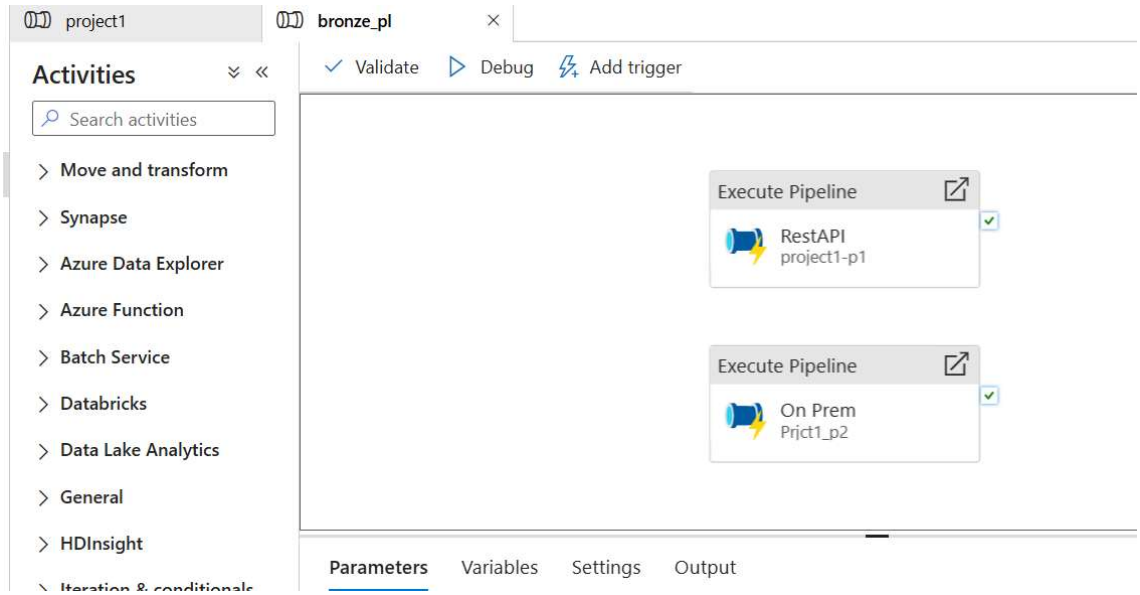

Flow:



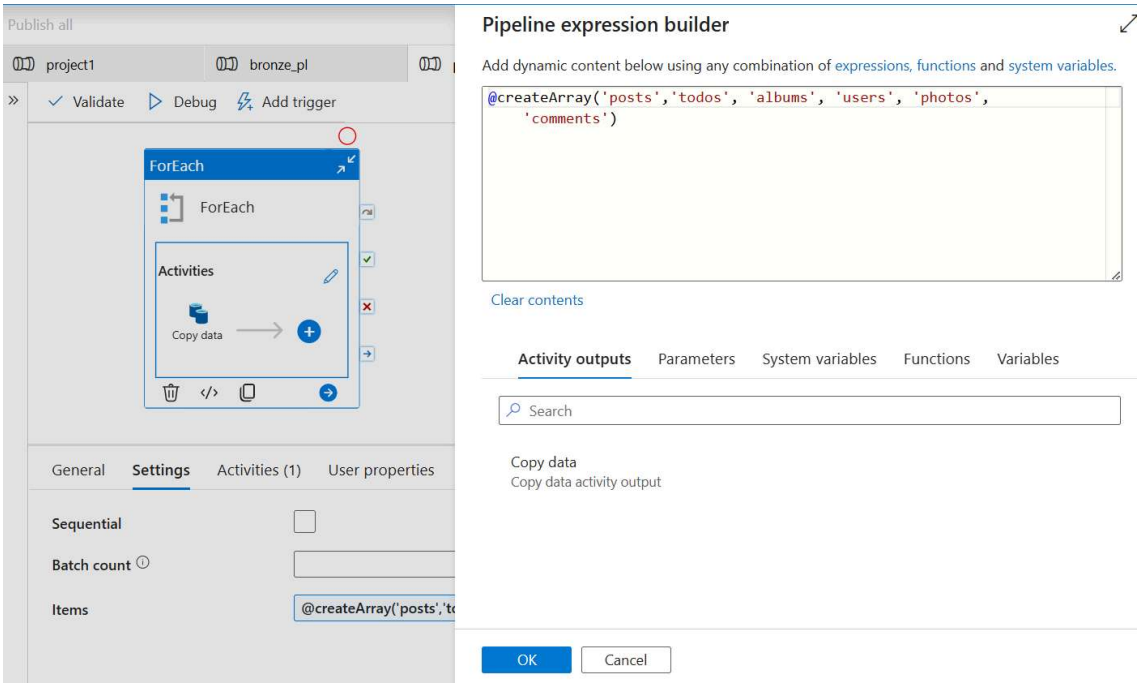
**Bronze layer(Execute pipeline)**

Rest API(Execute pipeline)<Rest API to SQL db(ADLSG2) logic

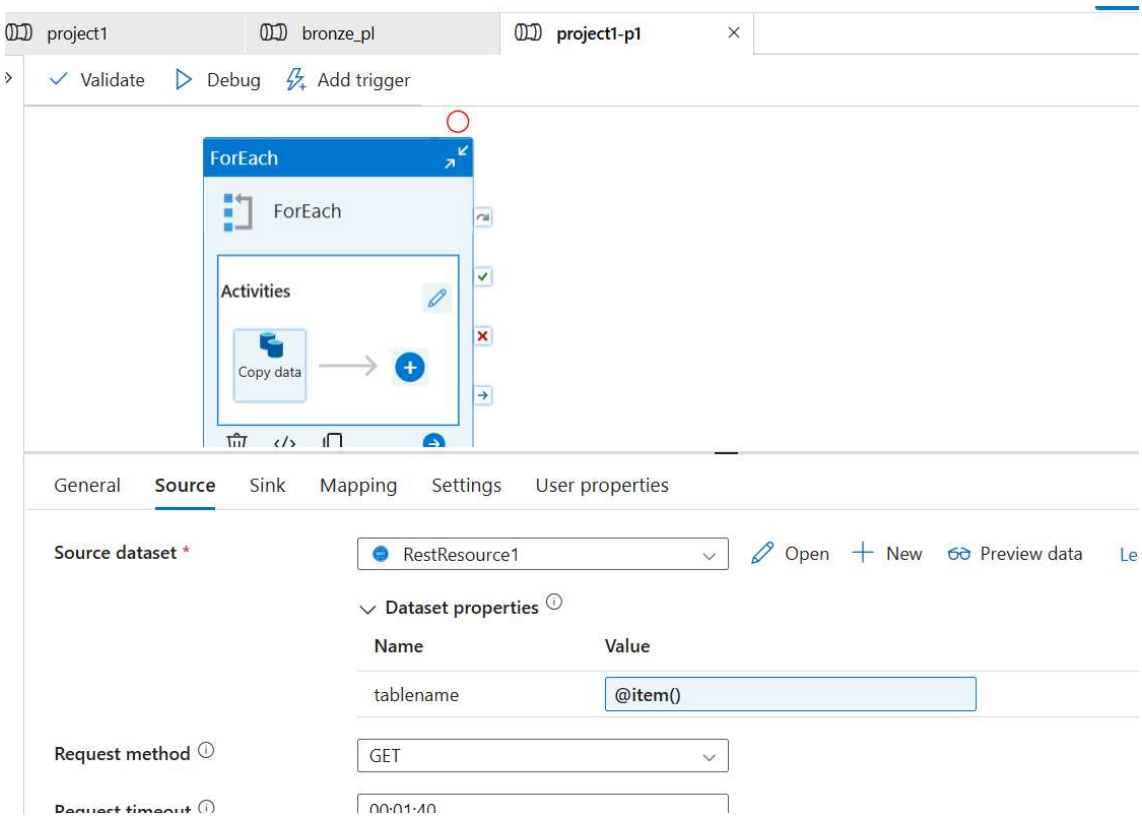
Onprem(Execute pipeline)<Incremental loading using metadata table-On prem to SQL db(ADLSG2) logic (parquet)



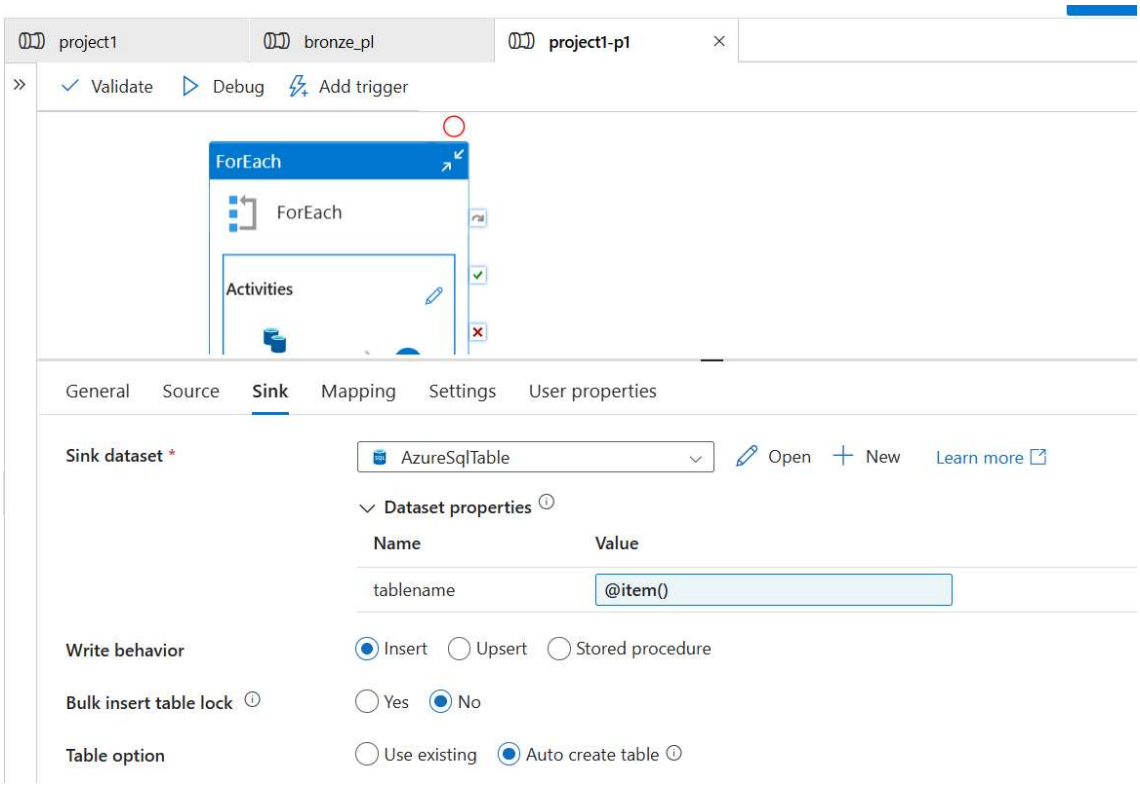

Part 1:
Rest API to SQL DB:
Passing array in the foreach.



Adding a copy activity inside the foreach with source as RestAPI:



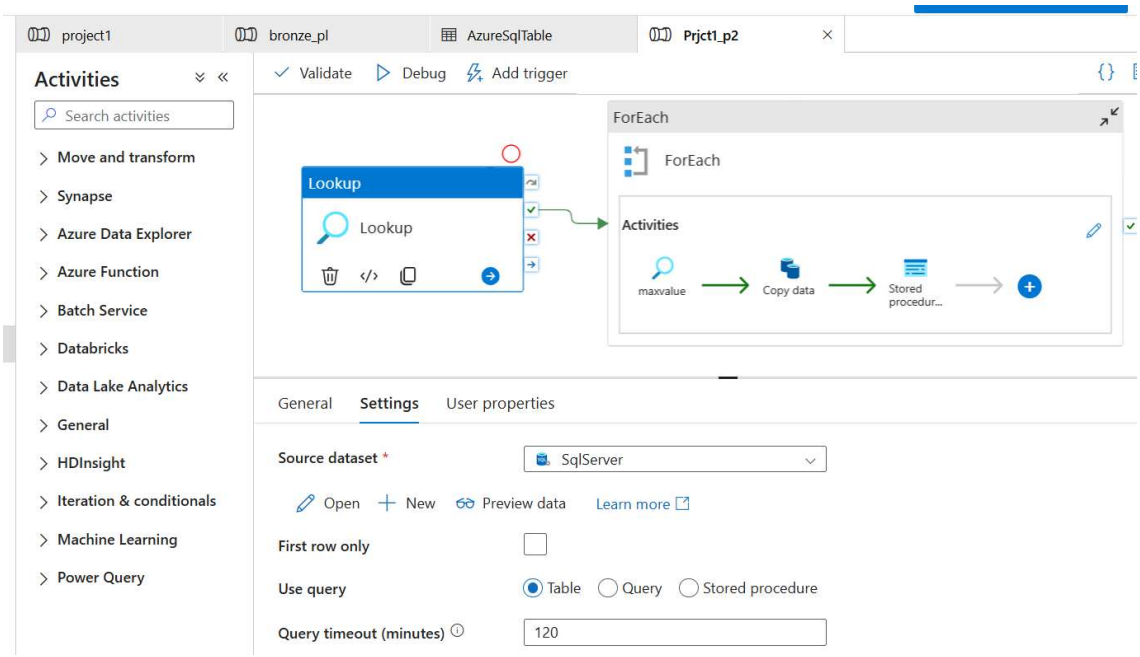

Sinking it to Azure sql database.



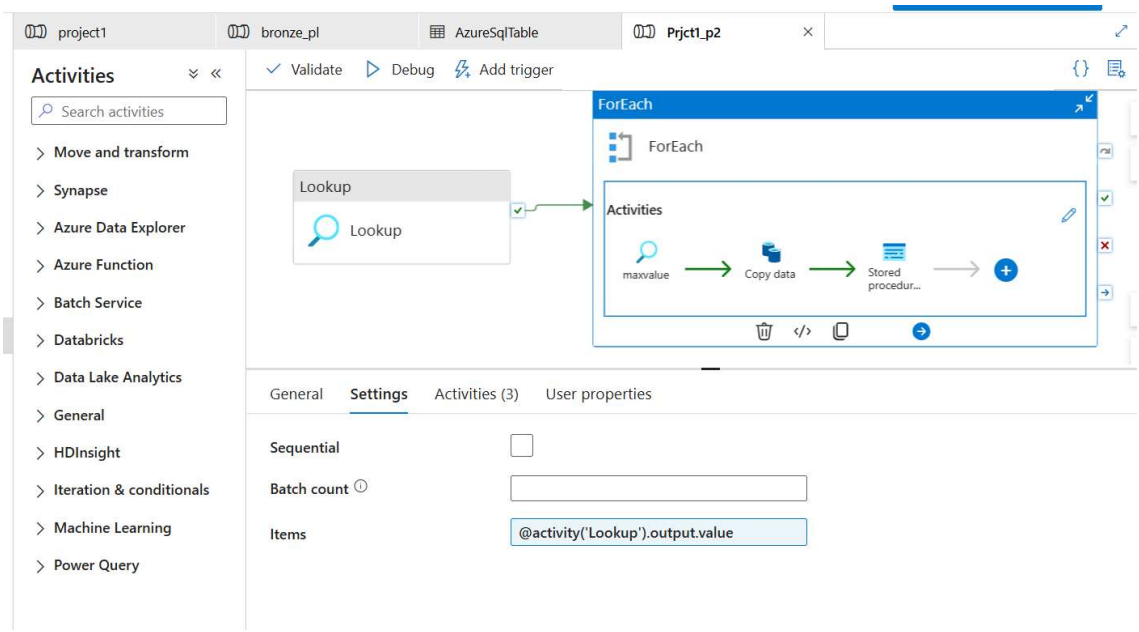
Part-2:

On prem to ADLSG2:

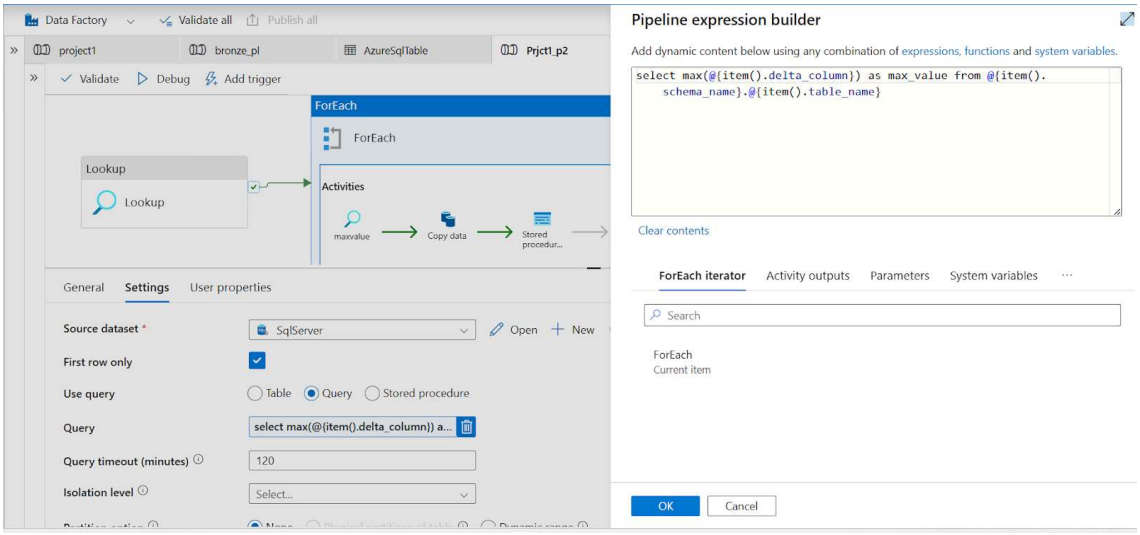
Looking up the data using look up activity.



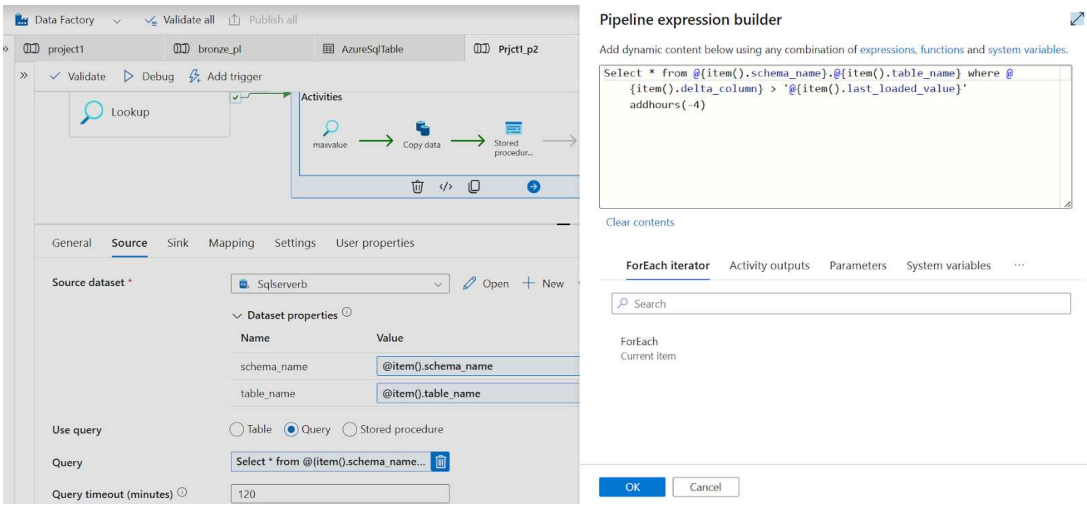

Passing its output value to foreach activity.



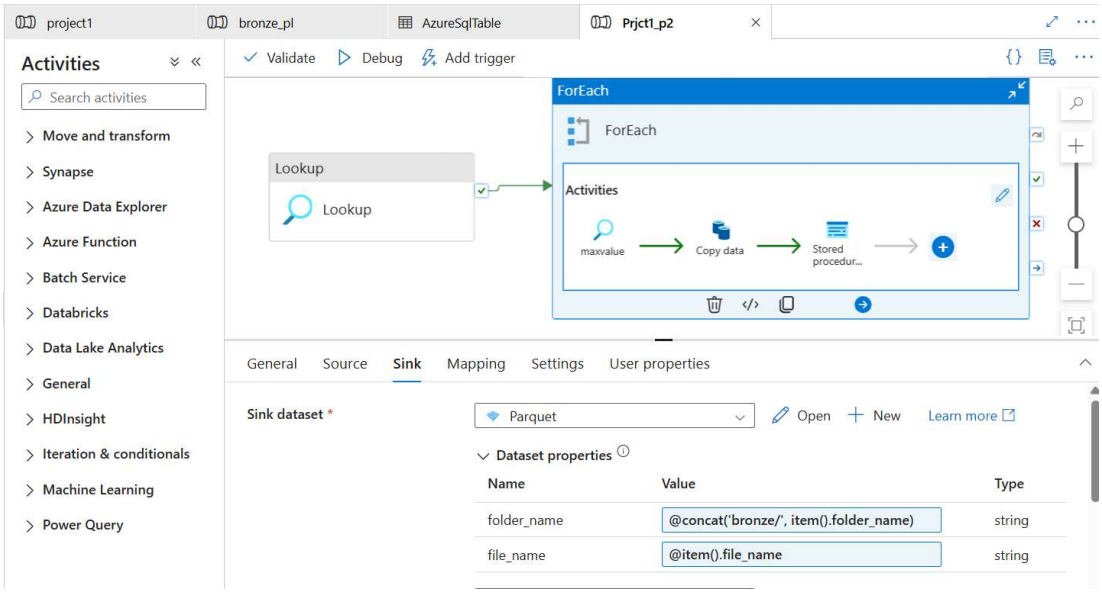
Adding a lookup activity to select max value from the delta column.



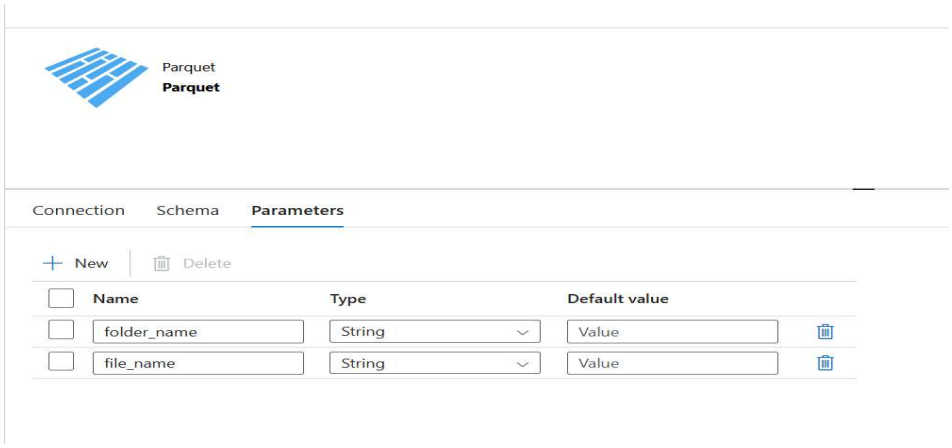

On success of max value look up, copy data. Source is on prem sql data, adding a select query where the delta column is greater than last loaded value. Adding negative hours for dealing with late arrival data.



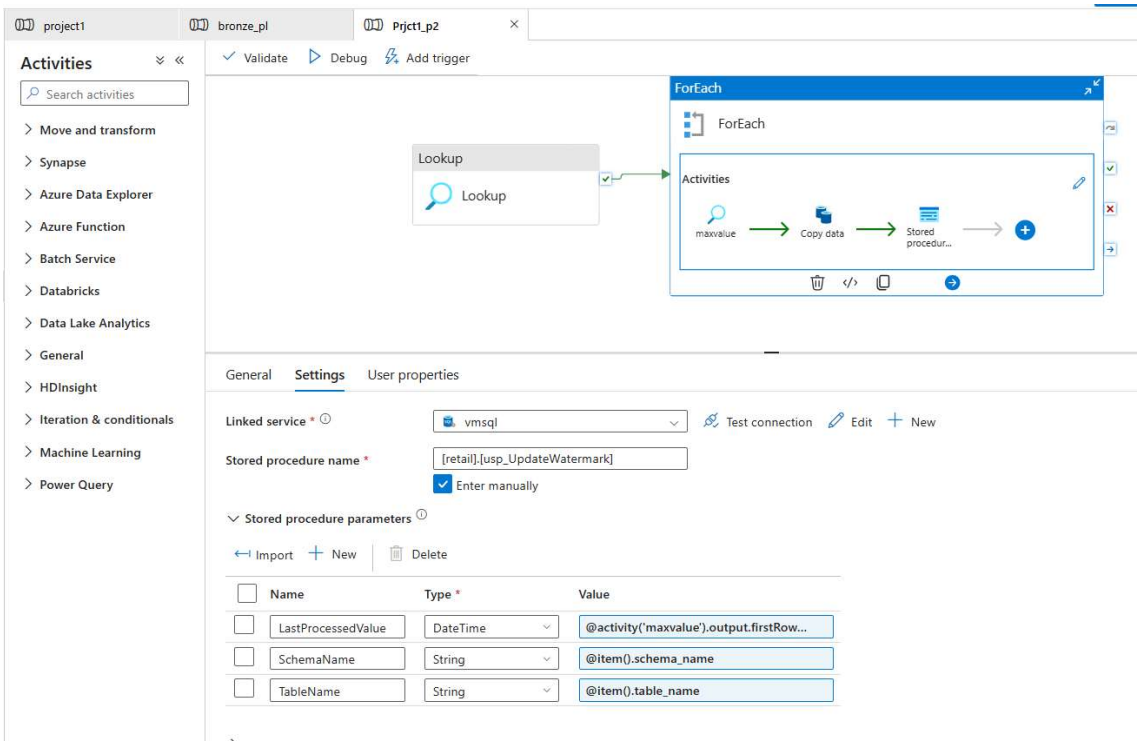
Sinking to the desired location i.e., bronze folder in parquet format.



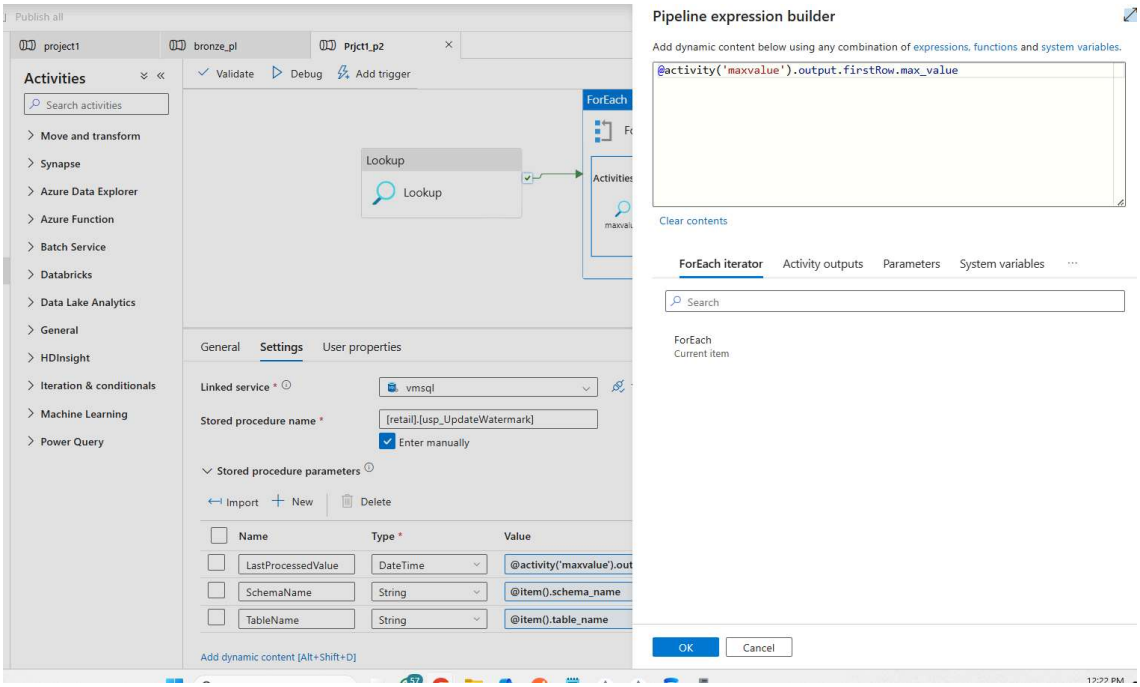

Adding parameters.



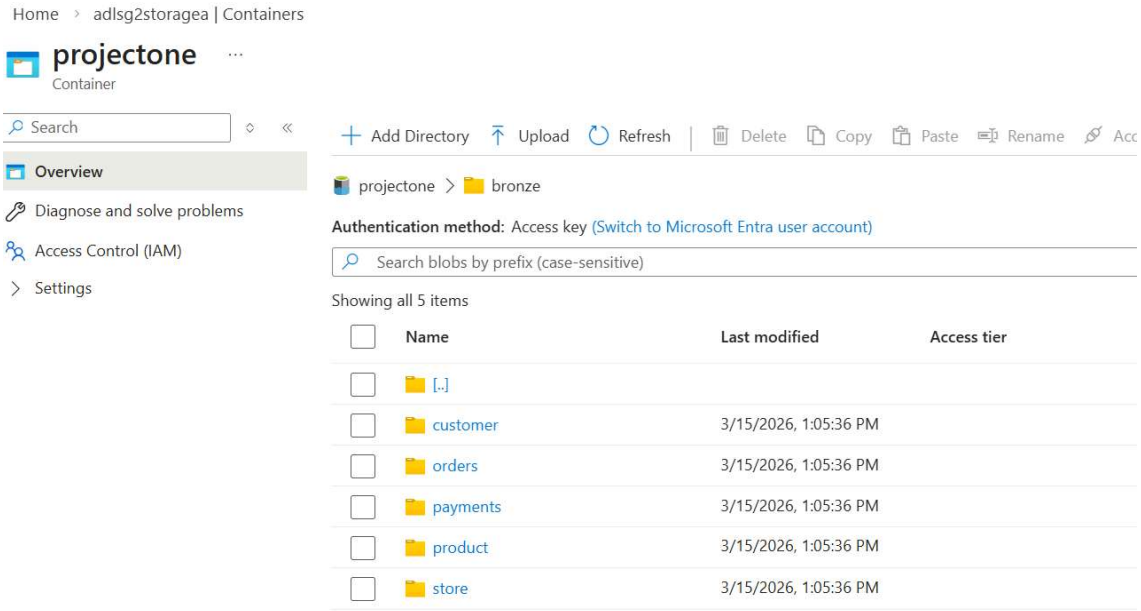
Stored procedure for dynamic autoloading:




Expression for last processed value:



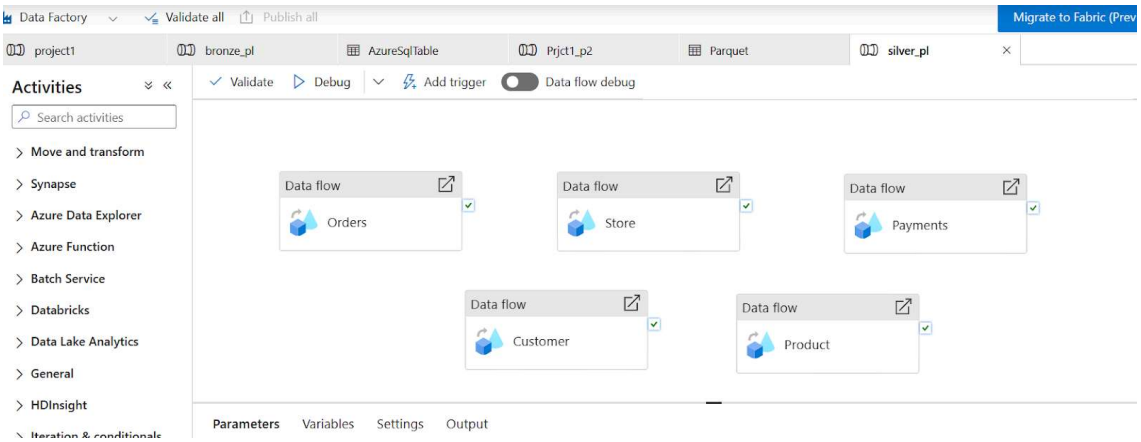
Result of the pipeline debug:



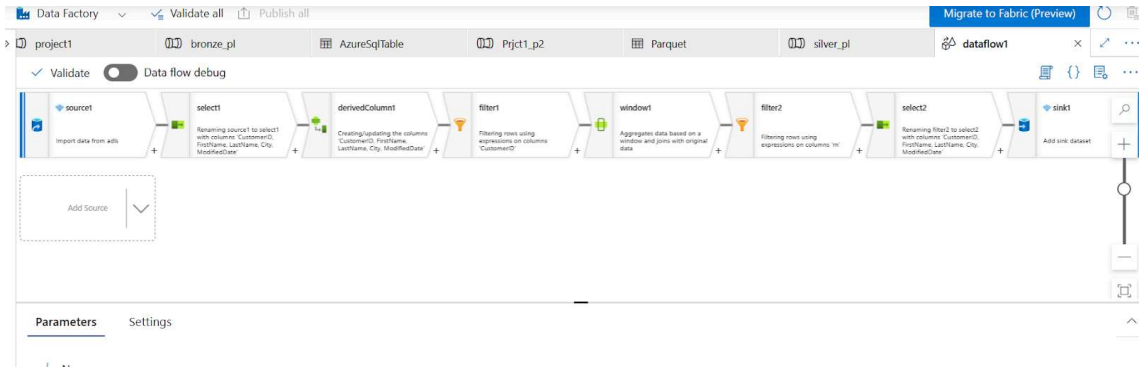

**Silver layer (Execute pipeline)**

Clean the data using data flow and store in ADLSG2 (parquet format)

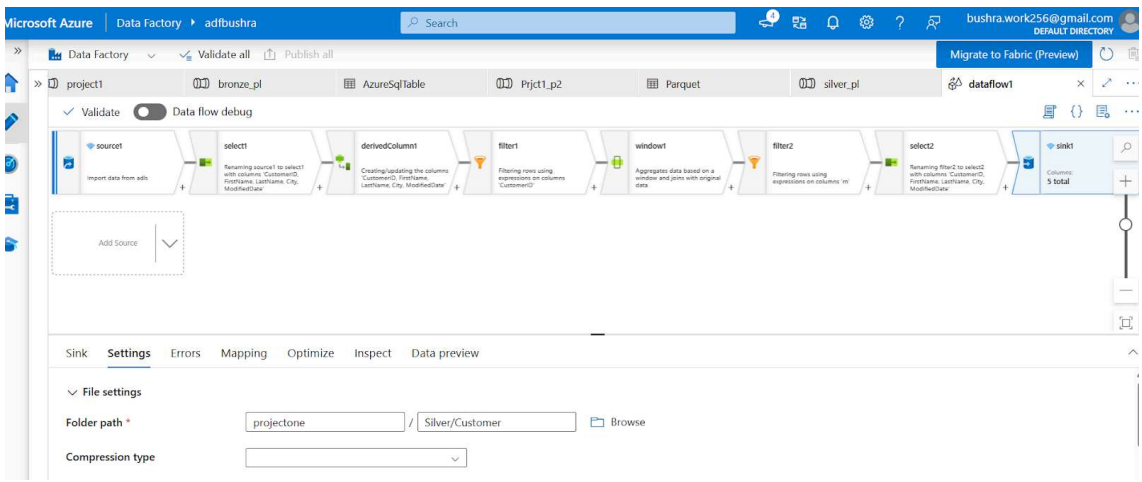
- Clean duplicates-Window transformation
- Check if primary key is null or zero
- Remove unnecessary columns-Select transformation
- Type casting-Derived column



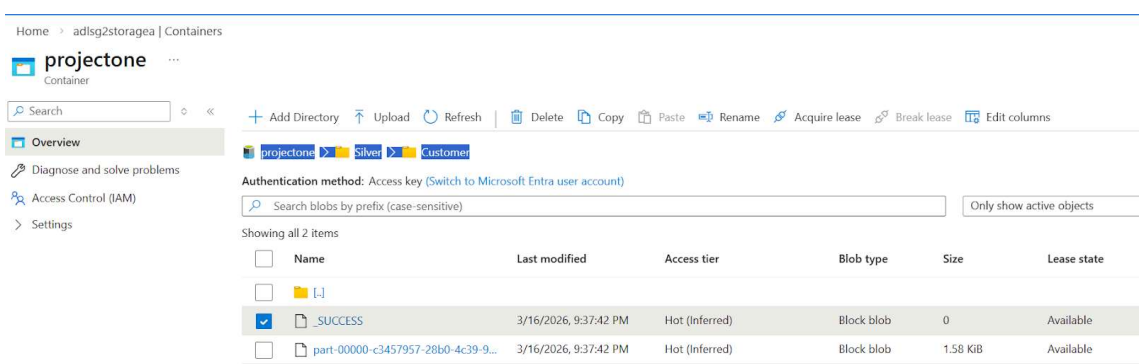
Cleaning and filtering the raw data from the bronze layer using data flows.




Sinking the cleaned and filtered data into the silver layer in parquet format.

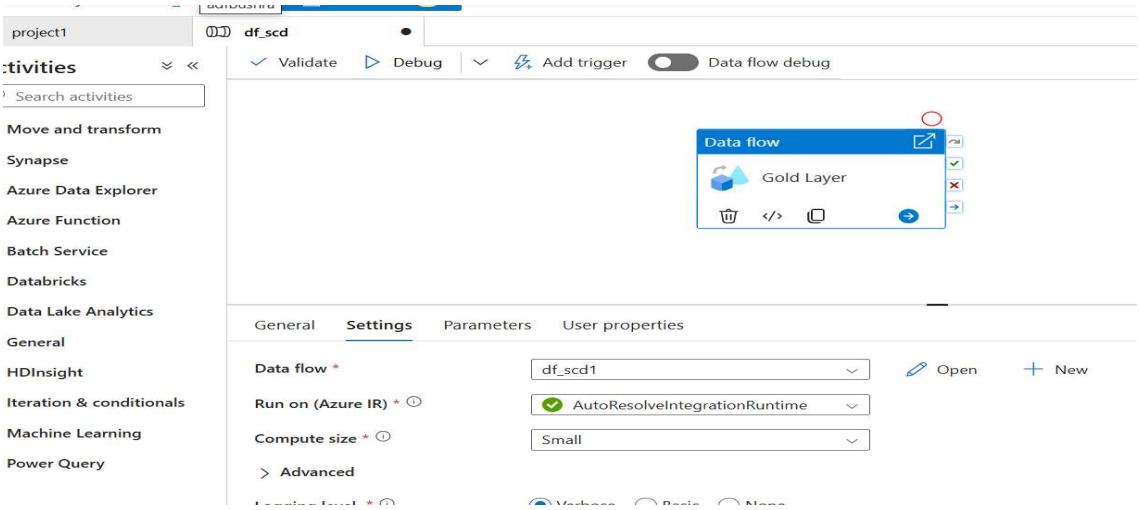



Result of the pipeline debug:

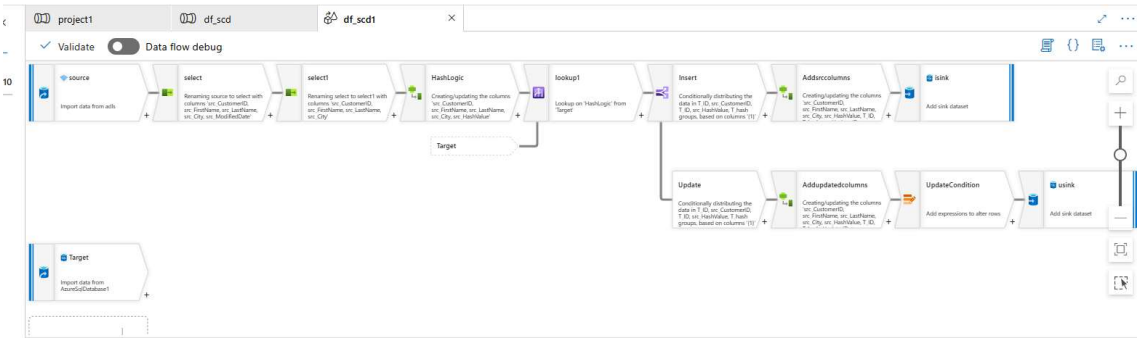



**Gold layer (Execute pipeline)**

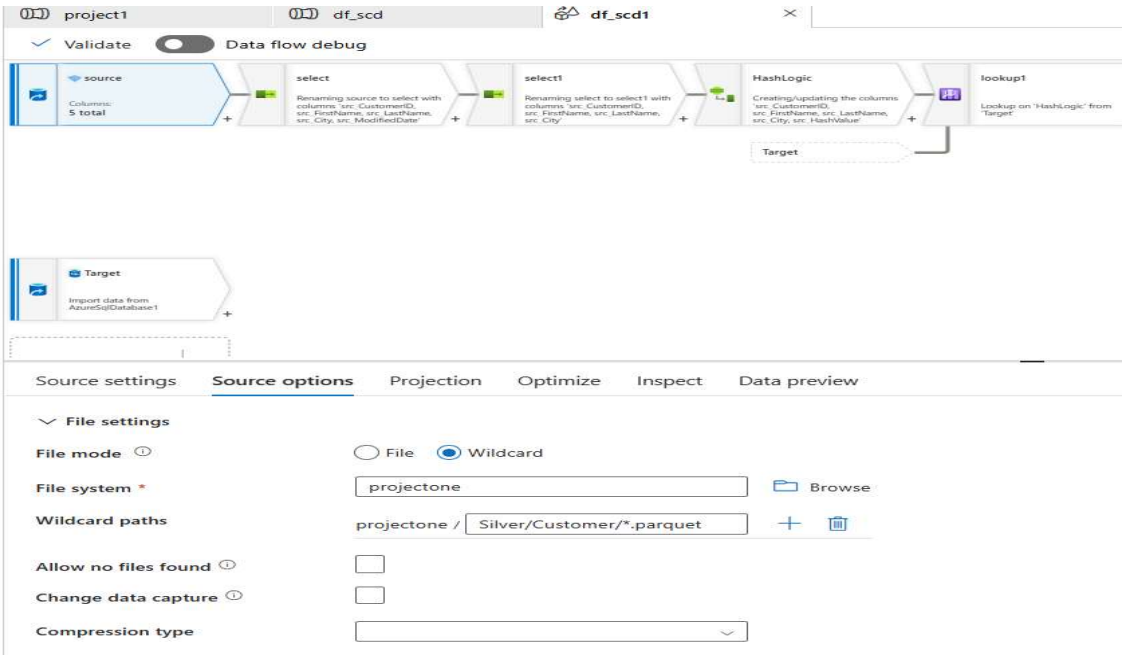
SCD Type 1 on one table from silver layer ADLSG2 to SQL database.



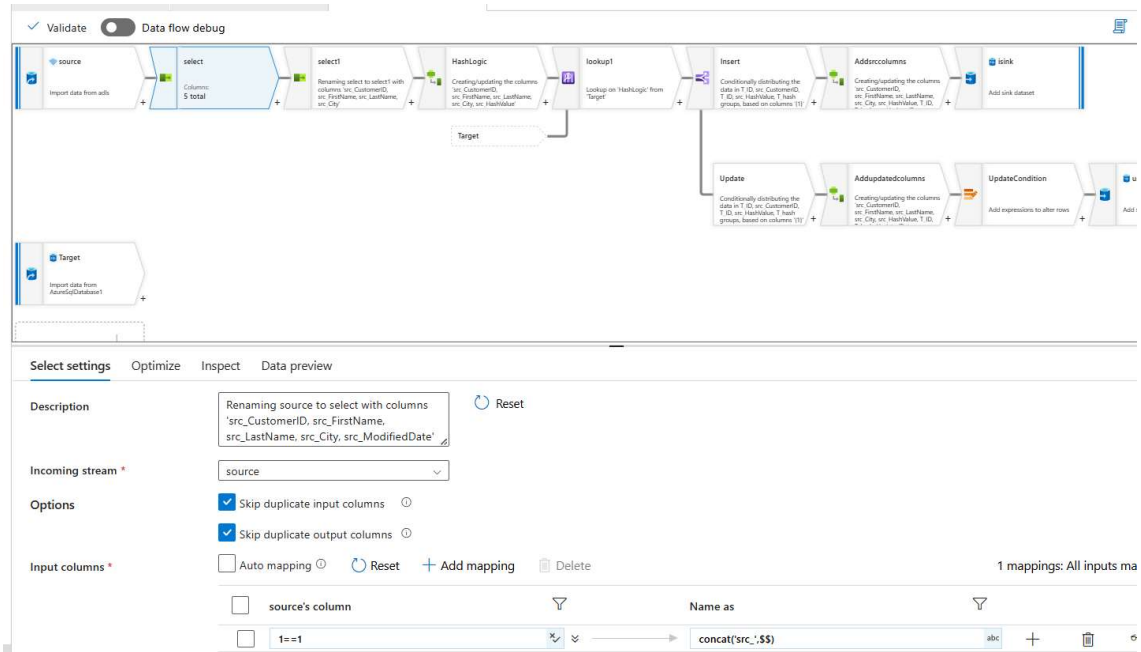

Performing SCD type 1in the gold layer using dataflow:



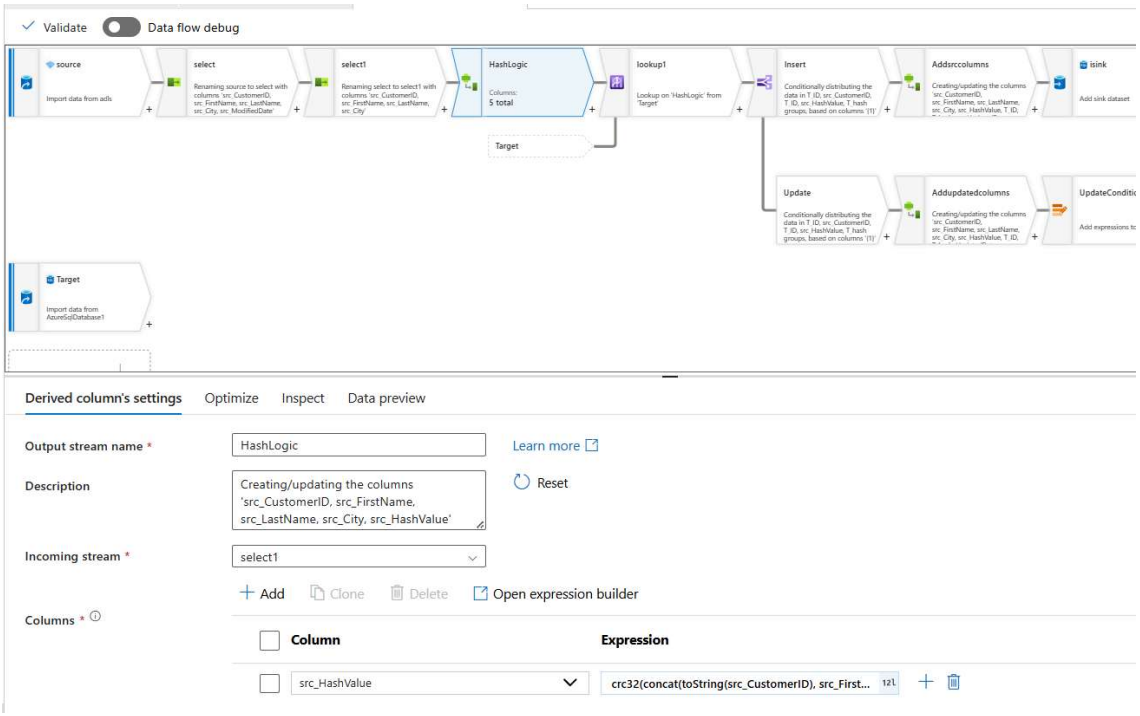

Customer data as source:



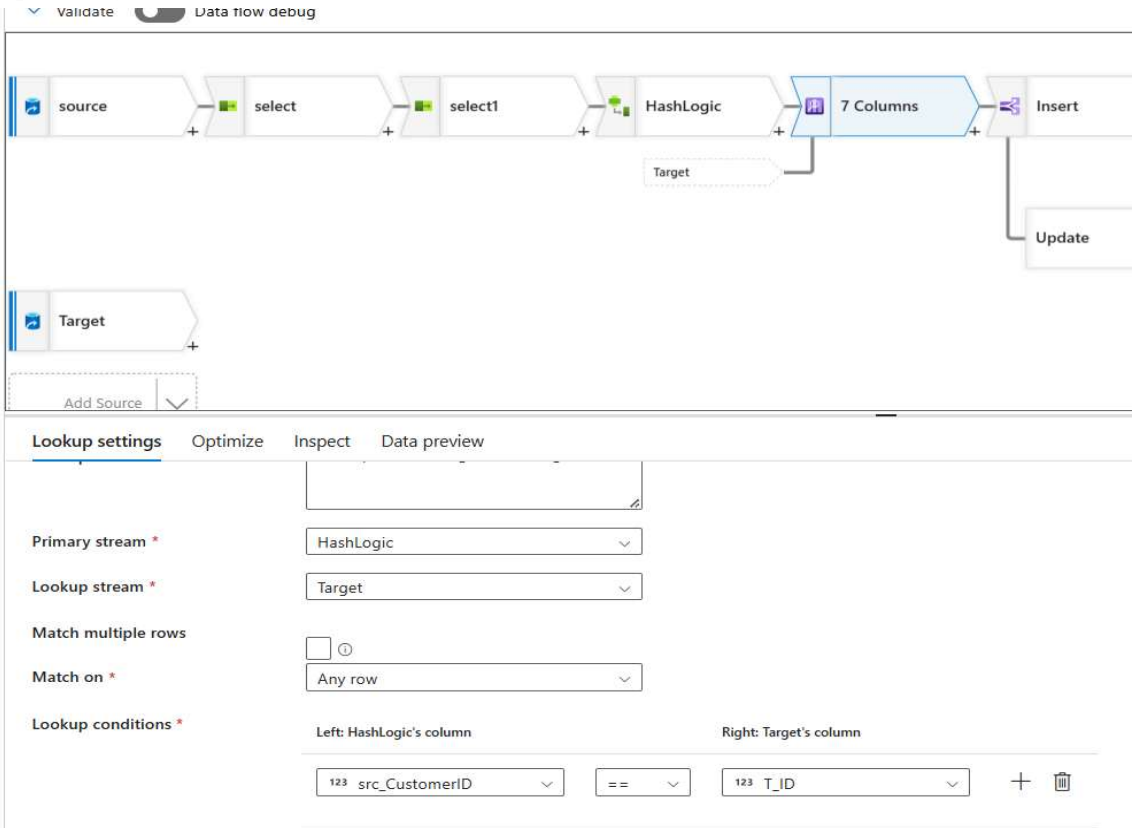
Appending all columns by src_



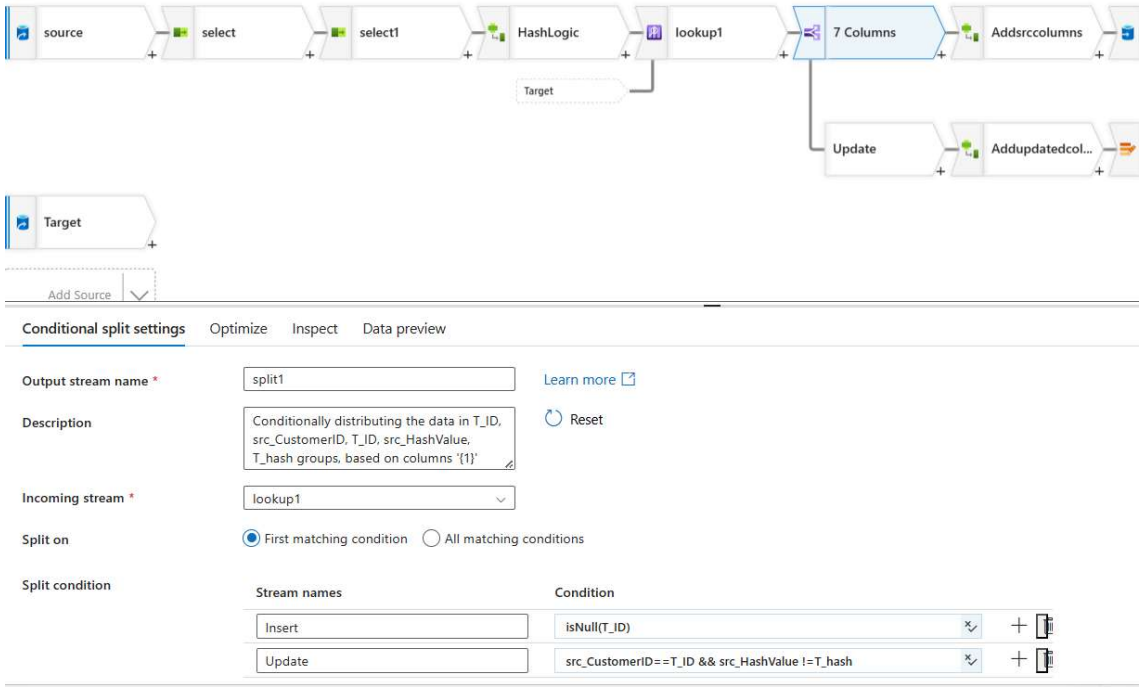

Adding a new column(hash column) and applying hashlogic using derived column transformation:



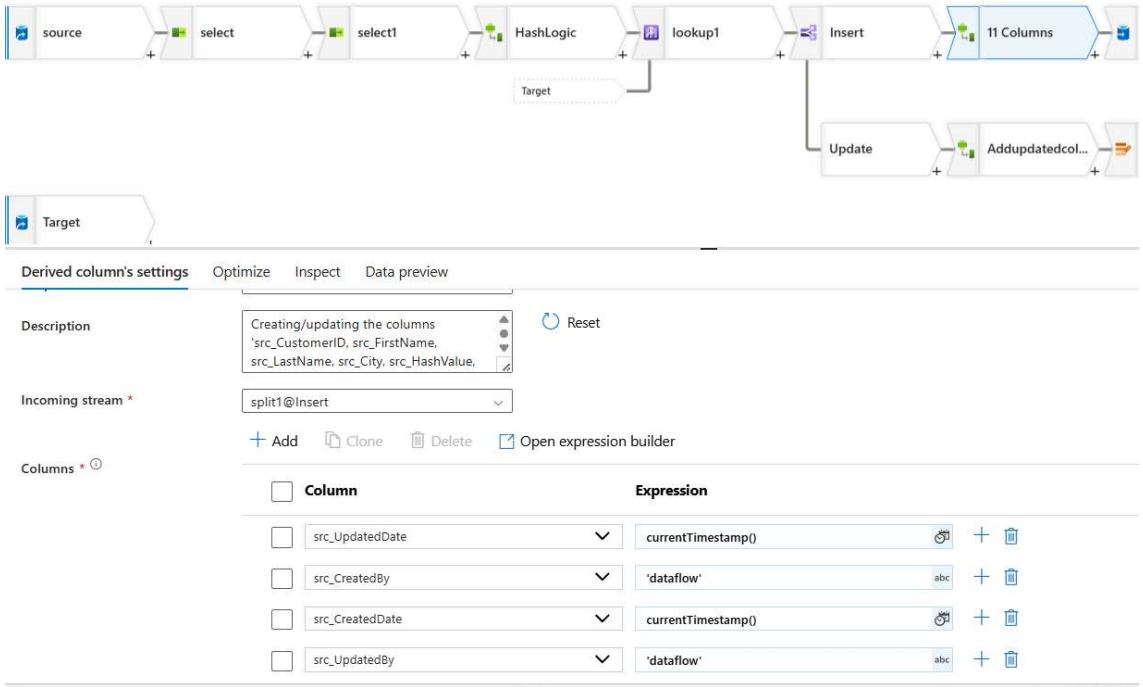
Add look up transformation and pass hashed table along with target stream and with source id equals target id condition:



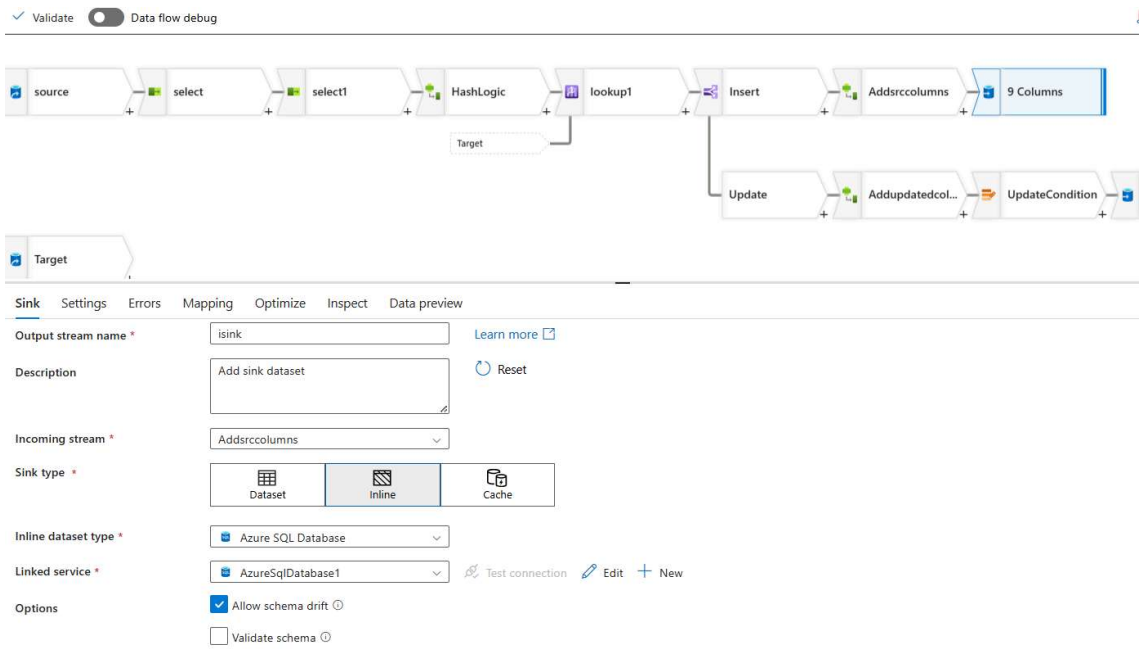

Split the resultant into insert and update streams with respective conditions:



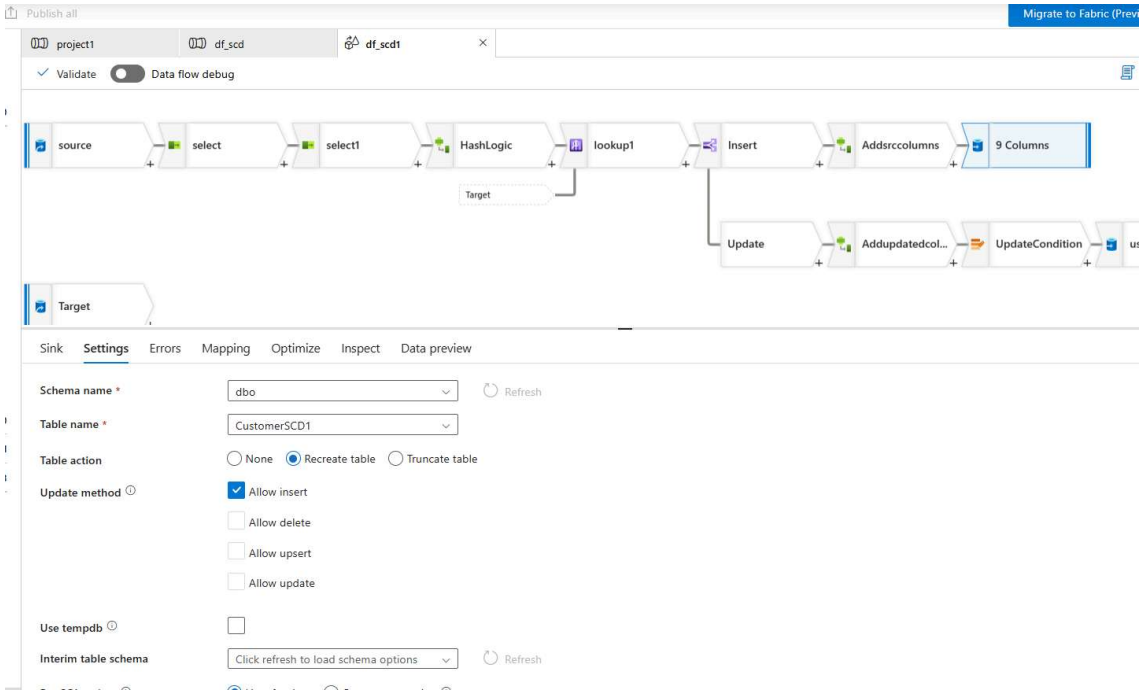
For insert stream, use a derived column transformation to add four new src_ columns as below.



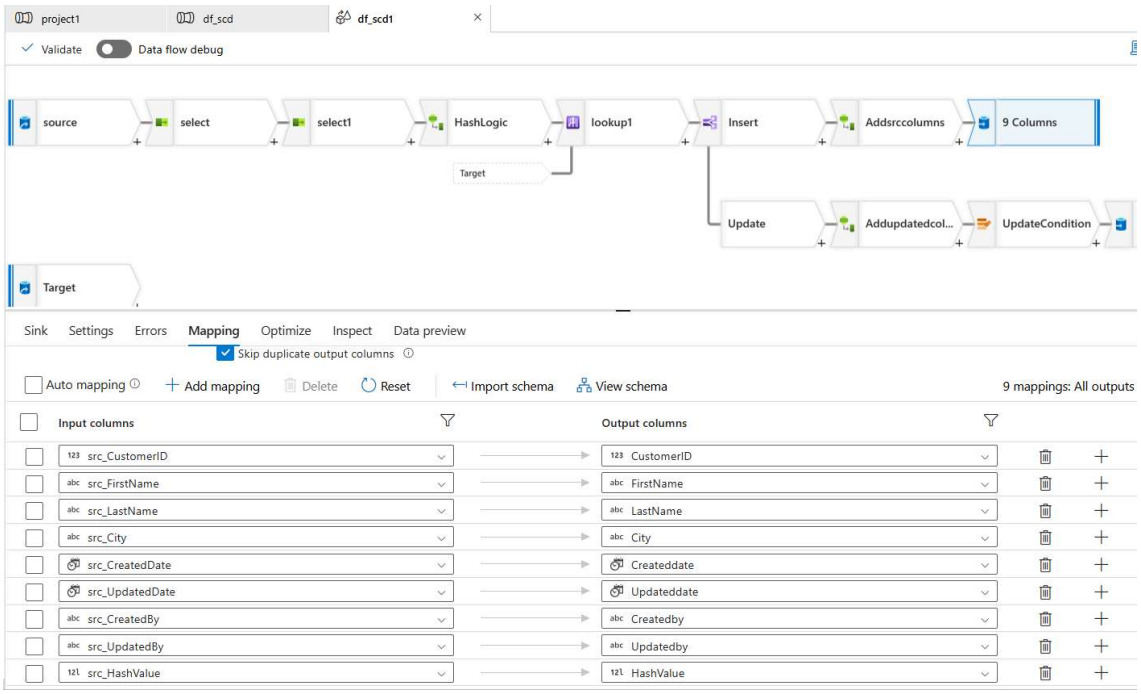

Sink the data into Azure sql database:



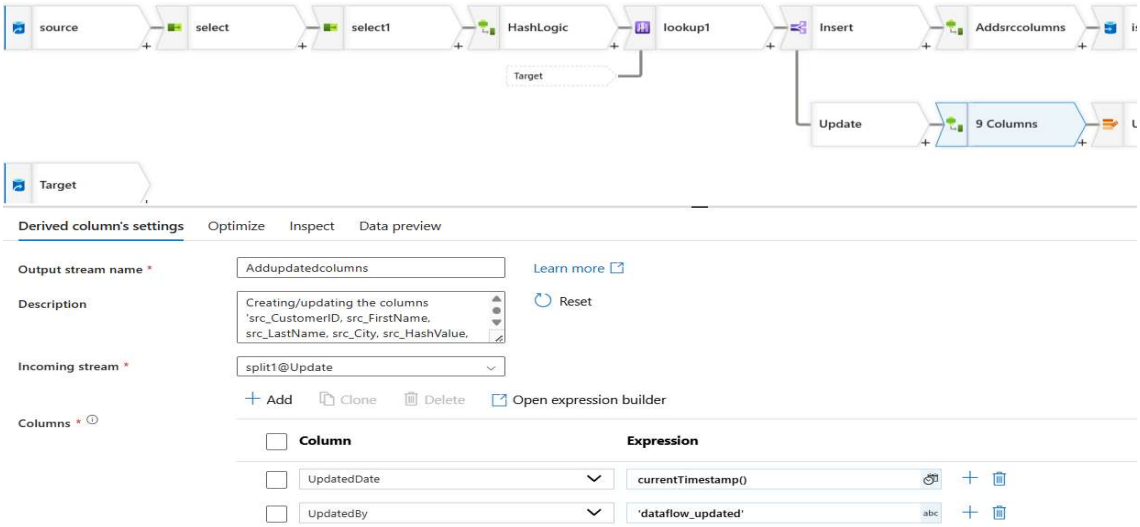
Select the target table and perform mapping:



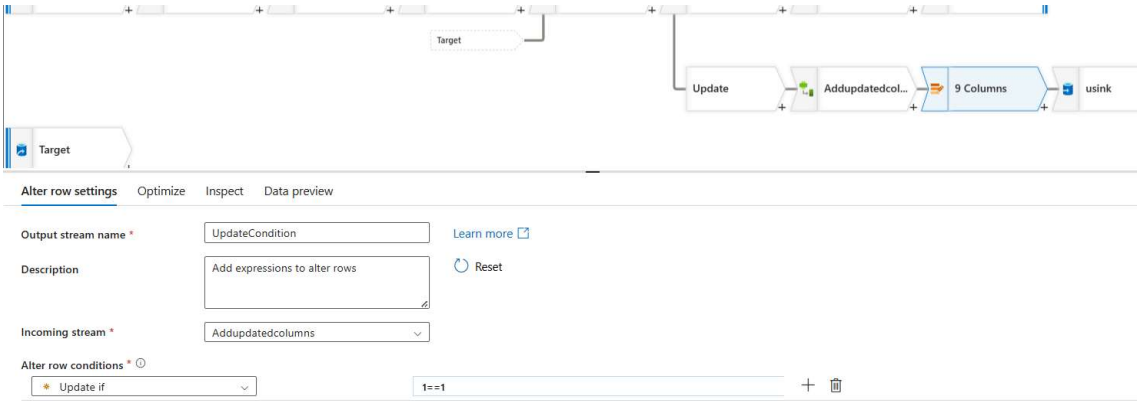

Mapping:



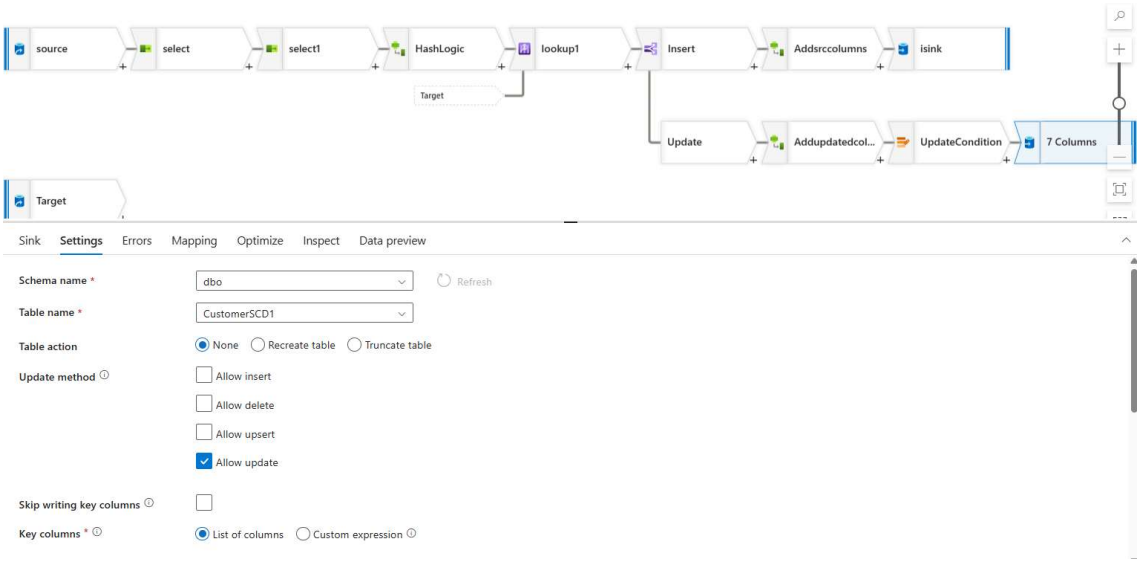
Now for update stream add two new columns as below using derived column transformation:



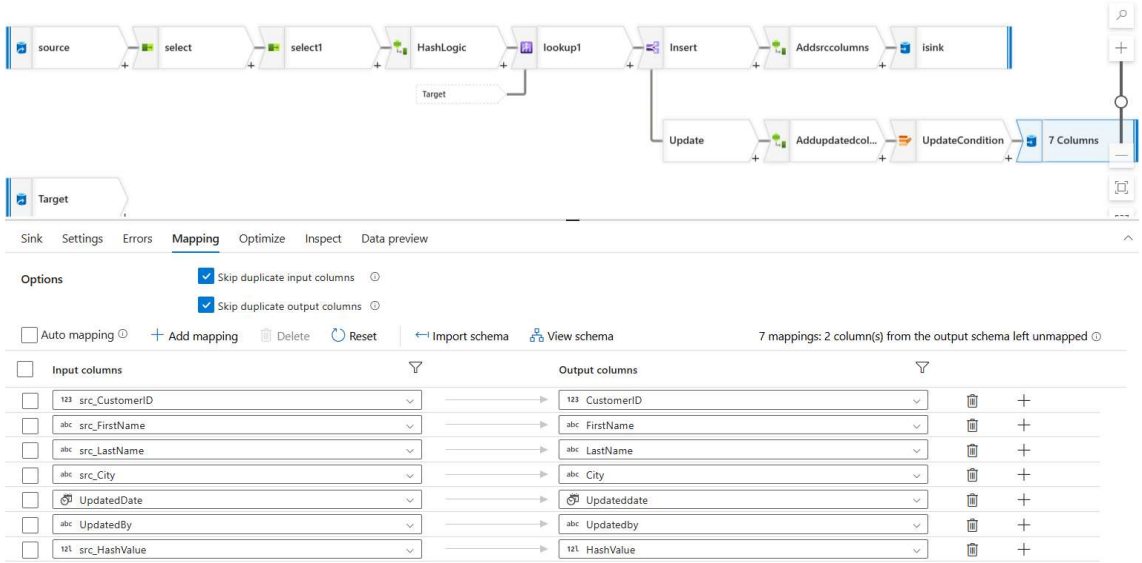

Add alter row condition:



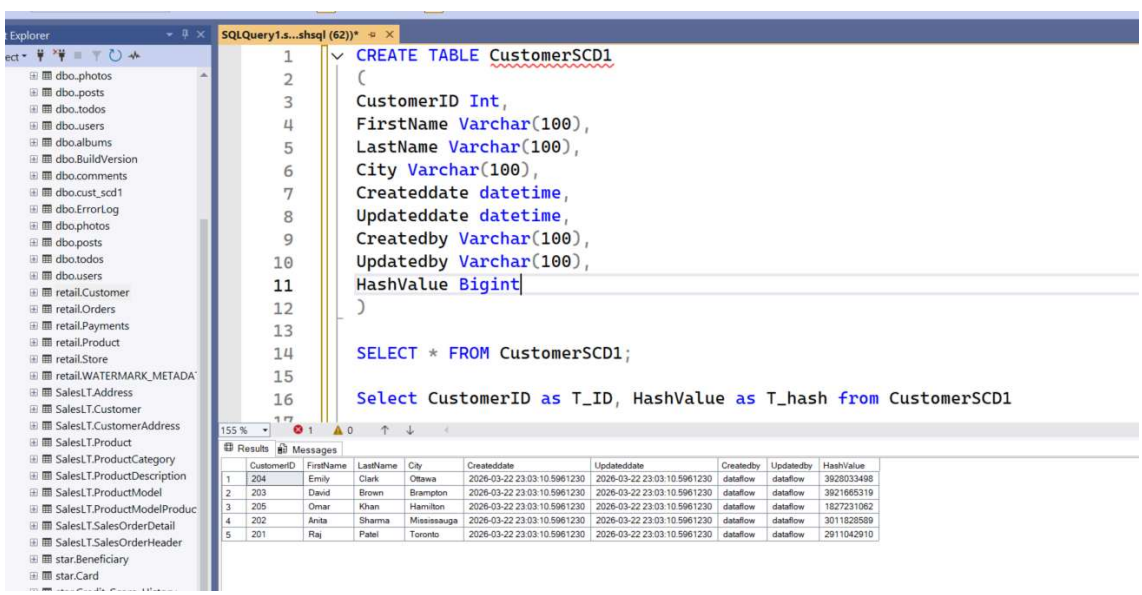

Add sink details [Allow update] for the target table and perform mapping:



Mapping:




Check the table in SSMS:




| | CustomerID | FirstName | LastName | City | Createddate | Updateddate | Createdby | Updatedby | HashValue |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 204 | Emily | Clark | Ottawa | 2026-03-22 23:03:10.5961230 | 2026-03-22 23:03:10.5961230 | dataflow | dataflow | 3928033498 |
| 2 | 203 | David | Brown | Brampton | 2026-03-22 23:03:10.5961230 | 2026-03-22 23:03:10.5961230 | dataflow | dataflow | 3921665319 |
| 3 | 205 | Omar | Khan | Hamilton | 2026-03-22 23:03:10.5961230 | 2026-03-22 23:03:10.5961230 | dataflow | dataflow | 1827231062 |
| 4 | 202 | Anita | Sharma | Mississauga | 2026-03-22 23:03:10.5961230 | 2026-03-22 23:03:10.5961230 | dataflow | dataflow | 3011828589 |
| 5 | 201 | Raj | Patel | Toronto | 2026-03-22 23:03:10.5961230 | 2026-03-22 23:03:10.5961230 | dataflow | dataflow | 2911042910 |